Module 39

Storage and File Structure/1: Physical Storage

Partha Pratim Das

Department of Computer Science and Engineering
Indian Institute of Technology, Kharagpur
ppd@cse.iitkgp.ac.in

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Module Recap

- Introduced Non-linear Data Structures - graph, tree, hash table
- Studied Binary Search Tree as an adaptation of binary search
- Compared Linear and Non-Linear Data Structures

Module Objectives

- Introduce various Physical Storage Media for high volume, fast, reliable and inexpensive options for data storage for databases
- To understand the options of Tertiary Storage for high volume, inexpensive backup options

Module Outline

- Physical Storage Media
- Magnetic Disks
- Magnetic Tape
- Other Storage
- Future of Storage

Physical Storage Media

Classification of Physical Storage Media

- Speed with which data can be accessed
- Cost per unit of data
- Reliability
  - data loss on power failure or system crash
  - physical failure of the storage device
- Can differentiate storage into:
  - **volatile storage** : loses contents when power is switched off
  - **non-volatile storage** : Contents persist even when power is switched off. Includes secondary and tertiary storage, as well as battery-backed up main-memory

Physical Storage Media

Cache

- fastest and most costly form of storage
- volatile
- managed by the computer system hardware

Main memory

- fast access (10's to 100's of nanoseconds)
- generally too small (or too expensive) to store the entire database
- capacities of up to a few Gigabytes widely used currently
- Capacities have gone up and per-byte costs have decreased steadily and rapidly

Volatile

contents of main memory are usually lost if a power failure or system crash occurs

Physical Storage Media (2): Flash memory

- Data survives power failure
- Data can be written at a location only once, but location can be erased and written to again
- Can support only a limited number (10K - 1M) of write/erase cycles
- Erasing of memory has to be done to an entire bank of memory
- Reads are roughly as fast as main memory
- But writes are slow (few microseconds), erase is slower
- Widely used in embedded devices such as digital cameras, phones, and USB keys

Physical Storage Media (3): Magnetic Disk

- Data is stored on spinning disk, and read/written magnetically
- Primary medium for the long-term storage of data
- typically stores entire database
- Data must be moved from disk to main memory for access, and written back for storage - much slower access than main memory
- direct-access
- possible to read data on disk in any order, unlike magnetic tape
- Capacities range up to roughly 16-32 TB
- Much larger capacity and much lower cost/byte than main memory/flash memory
- Growing constantly and rapidly with technology improvements
- Survives power failures and system crashes (disk failure can destroy data, but is rare)

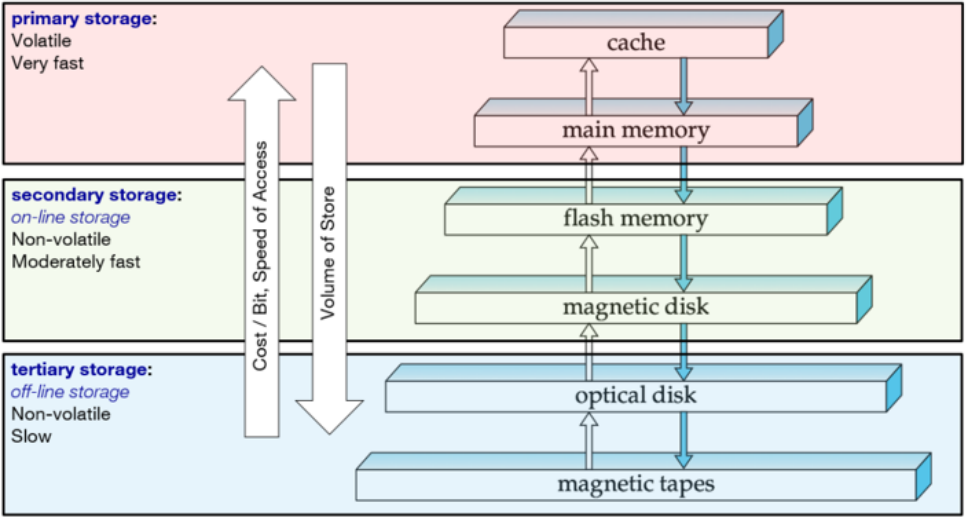
Physical Storage Media (4): Optical Storage

- non-volatile, data is read optically from a spinning disk using a laser
- CD-ROM (640 MB) and DVD (4.7 to 17 GB) most popular forms
- Blu-ray disks: 27 GB to 54 GB
- Write-one, read-many (WORM) optical disks used for archival storage (CD-R, DVD-R, DVD+R)
- Multiple write versions also available (CD-RW, DVD-RW, DVD+RW, and DVD-RAM)
- Reads and writes are slower than with magnetic disk
- Juke-box systems, with large numbers of removable disks, a few drives, and a mechanism for automatic loading/unloading of disks available for storing large volumes of data

Physical Storage Media (5): Tape Storage

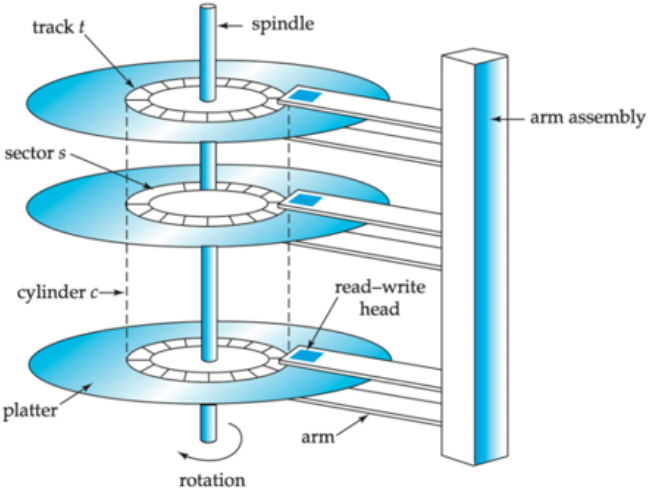
- non-volatile, used primarily for backup (to recover from disk failure), and for archival data
- sequential-access
- much slower than disk
- very high capacity (40 to 300 TB tapes available)
- tape can be removed from drive storage costs much cheaper than disk, but drives are expensive
- Tape jukeboxes available for storing massive amounts of data: hundreds of terabytes (TB) to multiple petabytes (PB)

Storage Hierarchy



Magnetic Disk

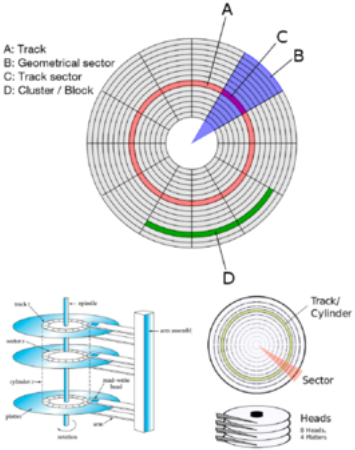
Magnetic Disk: Mechanism



NOTE: Diagram is schematic, and simplifies the structure of actual disk drives

Magnetic Disk (2): Mechanism

- Read-write head is positioned very close to the platter surface, reads/writes magnetically encoded information.
- Surface of platter divided into circular tracks (Over 50K-100K tracks per platter).
- Each track is divided into sectors. A sector is the smallest unit of data read or written (typically 512 bytes).
- To read/write a sector: disk arm swings to position head on right track, platter spins: Read/Write as sector passes under head.
- **Head-disk assemblies**: multiple disk platters on a single spindle (1 to 5 usually), one head per platter, mounted on a common arm.
- **Cylinder** i consists of i-th track of all the platters



Magnetic Disks (3): Disk Controller, Subsystems, and Interfaces

- **Disk Controller** : interfaces between the computer system and the disk drive hardware
  - Accepts high-level commands to read or write a sector
  - Initiates actions moving the disk arm to the right track, reading or writing the data
  - Computes and attaches checksums to each sector to verify that correct read back
  - Ensures successful writing by reading back sector after writing it
  - Performs remapping of bad sectors
- **Disk Subsystem** :
  - Disk Interface Standards Families : ATA, SATA, SCSI, SAS, several variants
  - Storage Area Networks (SAN) connects disks by a high-speed network to a number of servers
  - Network Attached Storage (NAS) provides a file system interface using networked file system protocol

Magnetic Disks (4): Performance Measures

- **Access Time** : time from a read or write request issue to start of data transfer:
  - **Seek Time** : time to reposition the arm over the correct track (4 to 10 ms)
  - **Rotational Latency** : time for the sector to be accessed to appear under the head (4 to 11 ms)
- **Data-transfer Rate** : the rate at which data can be retrieved from or stored to the disk (25 to 100 MB per second max rate)
- **Mean Time To Failure (MTTF)** : Avg. time the disk is expected to run continuously without any failure
  - Typically 3 to 5 years (theoretical MTTF of 500,000 to 1,200,000 hours)
  - MTTF decreases as disk ages

Magnetic Tapes

Magnetic Tapes

- Hold large volumes of data and provide high transfer rates
- Tape Formats: DAT, DLT, Ultrium, Ampex helical scan format
- Transfer rates from few to 10's of MB/s
- Tapes are cheap, but cost of drives is very high
- Very slow access time in comparison to magnetic and optical disks (Limited to sequential access)
- Used mainly for backup, for storage of infrequently used information, and as an off-line medium
- Tape jukeboxes used for very large capacity storage (Multiple petabyes)

Cloud Storage

Cloud Storage

- Cloud storage is purchased from a third-party cloud vendor who owns and operates data storage capacity and delivers it over the Internet in a pay-as-you-go model
- These cloud storage vendors manage capacity, security and durability to make data accessible to applications all around the world
- Applications access cloud storage through traditional storage protocols or directly via an API
- Many vendors offer complementary services designed to help collect, manage, secure and analyze data at massive scale. Various available options for cloud storage are:
  - Google Drive, Amazon Drive, OneDrive, Evernote, Dropbox, etc.

Cloud Storage vs. Traditional Storage

Parameters	Cloud Storage	Traditional Storage
Cost	Cloud storage is cheaper per GB than using external drives.	The hardware and infrastructure costs are high.
Reliability	Cloud storage is highly reliable.	Traditional storage requires high initial effort and is less reliable.
File Sharing	Supports file sharing dynamically as it can be shared anywhere with network access.	Requires physical drives to share data and a network is to be established between both.
Accessibility	Gives you access to your files from anywhere.	Restricted to local access.
Backup/ Recovery	Very safe from on site disaster. Acts as a backup solution for your local storage.	Data that is stored locally is much more susceptible to unexpected events and could be easily lost.

Other Storage

Optical Disks

Compact disk-read only memory (CD-ROM)

- Removable disks, 640 MB per disk
- Seek time about 100 msec (optical read head is heavier and slower)
- Higher latency (3000 RPM) and lower data-transfer rates (3-6 MB/s) compared to magnetic disks

Digital Video Disk (DVD) & Blu-ray

- DVD-5 holds 4.7 GB, and DVD-9 holds 8.5 GB
- DVD-10 and DVD-18 are double sided formats with capacities of 9.4 GB and 17 GB
- Blu-ray DVD: 27 GB (54 GB for double sided disk)
- Record once versions (CD-R and DVD-R) are popular (high capacity and long lifetime; archival)
- Multi-write versions (CD-RW, DVD-RW, DVD+RW and DVD-RAM) also available

Flash Drives

- Flash drives are often referred to as pen drives, thumb drives, or jump drives. They have completely replaced floppy drives for portable storage.
- They have nearly replaced CDs and DVDs for data storage purposes.
- USB flash drives are removable and rewritable storage devices that require a USB port for connection and utilizes non-volatile flash memory technology.
- The storage space in USB flash drives is quite large with sizes ranging from 128MB to 2TB.
- The USB standard a flash drive is built around will determine the number of things about its potential performance, including maximum transfer rate.

Secure Digital Cards (SD cards)

- A Secure Digital (SD, in short) card is a type of removable memory card used to read and write large quantities of data.
- Widely used in mobile electronics, cameras, smart devices, video game consoles.

Card Type	Year	Capacity	Supported Devices
SD	1996	128MB to 2GB	All host devices that support SD, SDHC, SDXC
SDHC	2006	4GB to 32GB	All host devices that support SDHC, SDXC
SDXC	2009	64GB to 2TB	All host devices that support SDXC

Card Type	Capacity	File System	Remarks
SD	128MB to 2GB	FAT16	FAT16 supports 16 MB to 2 GB
SDHC	4GB to 32GB	FAT32	FAT32 can be support up to 16 TB
SDXC	64GB to 2TB	exFAT	exFAT is non-standard, supports file up to 4 GB

Flash Storage

- NOR flash vs NAND flash
- **NAND flash**:
  - used widely for storage, since it is much cheaper than NOR flash
  - requires page-at-a-time read (page: 512 bytes to 4 KB)
  - transfer rate around 20 MB/sec
- **solid state disks** : Use multiple flash storage devices to provide higher transfer rate of 100 to 200 MB/sec
- erase is very slow (1 to 2 ms)
  - erase block contains multiple pages
  - remapping of logical page addresses to physical page addresses avoids waiting for erase (translation table tracks mapping)
  - after 100,000 to 1,000,000 erases, erase block becomes unreliable (wear leveling)

Solid-State Drives (SSD)

- SSDs replace traditional mechanical hard disks by using flash-based memory, which is significantly faster.
- SSDs speed up computers significantly due to their low read-access times and fast throughput.
- The idea of SSDs was introduced in 1978. It stores the data in the persistent state even when no power is supplied.
- The speed of SSD is much larger than that of HDD as it reads/writes data at higher input-output per second.
- Unlike HDDs, SSDs do not include any moving parts. SDDs can resist vibrations and high temperatures.

SDD vs. HDD

Parameters	SSD	HDD
Technology	Integrated circuit using Flash memory	Mechanical Parts, including spinning disks
Access Time	0.1 ms	5.5-8.0 ms
Average Seek Time	0.08-0.16 ms	$<$ 10 ms
Speed (SATA II)	80-250 MB/sec	65-85 MB/sec
Random I/O Performance	6000 io/s	400 io/s
Backup rates	6 hours	20-24 hours
Reliability	The failure rate of less than 0.5%	Failure rate fluctuates between 2-5%
Energy Consumption	2 to 5 watts	6 to 15 watts

Future of Storage

Future of Storage: DNA Digital Storage

- DNA digital data storage is the process of encoding and decoding binary data to and from synthesized strands of DNA.
- While DNA has enormous potential because of its high storage density, its practical use is currently severely limited because of its high cost and very slow read and write times.
- This information can be encoded in DNA using the four nucleotides: A, T, G, and C. For example, G and C could define 0 while A and T represent 1.
- DNA is extremely stable and is fairly easy (but expensive) to synthesize and sequence.
- High density - each nucleotide is about 1 cubic nanometer - an exabyte (10^{18} bytes) of data stored as DNA could fit in the palm of your hand.
- DNA Synthesis: A DNA synthesizer machine builds synthetic DNA strands matching the sequence of digital code.

Future of Storage: Quantum Memory

- Quantum memory is the quantum-mechanical version of ordinary computer memory.
- Whereas ordinary memory stores information as binary states, quantum memory stores a quantum state for later retrieval.
- These states hold useful computational information known as qubits.
- Quantum memory is essential for the development of many devices in quantum information processing applications such as quantum network, quantum repeater, linear optical quantum computation or long-distance quantum communication.
- Unlike classical memory, the states stored in quantum memory can be in a quantum superposition, giving much more practical flexibility in quantum algorithms.

Module Summary

- Understood the range of Physical Storage Media
- Studied the mechanism and performance of the Magnetic Disks
- Looked at the features of Magnetic Tape as tertiary storage
- Glimpsed through Other Storage including Optical Disk, Flash and SSD
- Considered the Future of Storage in terms of DNA and Quantum

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